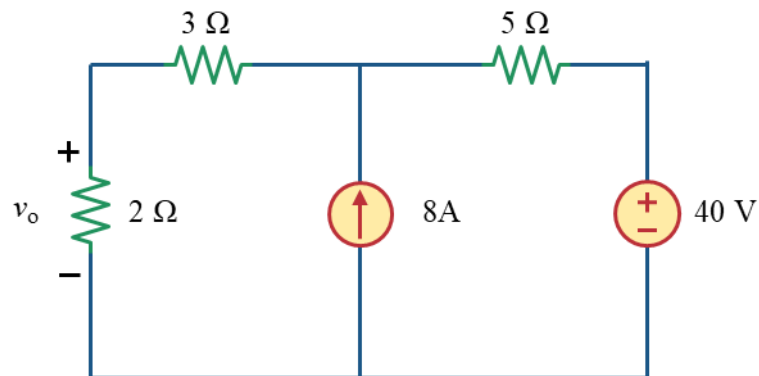


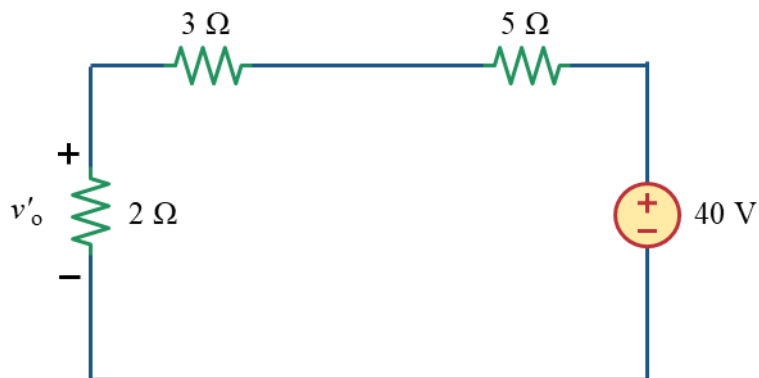
Solution to Problem 1



If v_o' and v_o'' are the contributions from the 40 V voltage source and 8 A current source respectively, then

$$v_o = v_o' + v_o''$$

First, Considering the contribution of the 40 V voltage source (replacing the current source (8 A) by an open circuit)

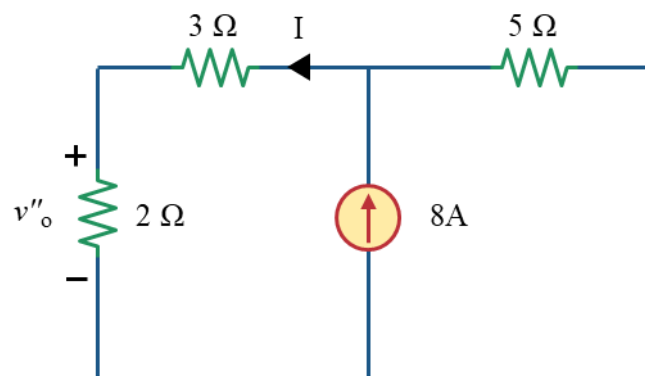


Using voltage divider rule,

$$v_o' = 40 \times \frac{2}{2 + 3 + 5}$$

$$v_o' = 8 \text{ V}$$

Considering the contribution of the 8 A current source (replacing the voltage source (40 V) by a short circuit)



Applying current divider rule,

$$I = 8 \times \frac{(3 + 2) \parallel 5}{3 + 2}$$

$$I = 4 \text{ A}$$

Now,

$$v_0'' = I \times 2$$

$$v_0'' = 4 \times 2$$

$$v_0'' = 8 \text{ V}$$

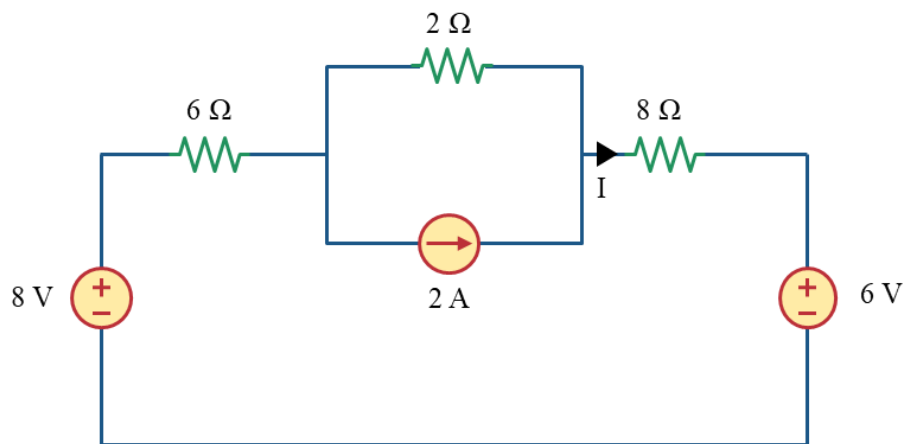
Finally,

$$v_0 = v_0' + v_0''$$

$$v_0 = 8 + 8$$

$$v_0 = 16 \text{ V}$$

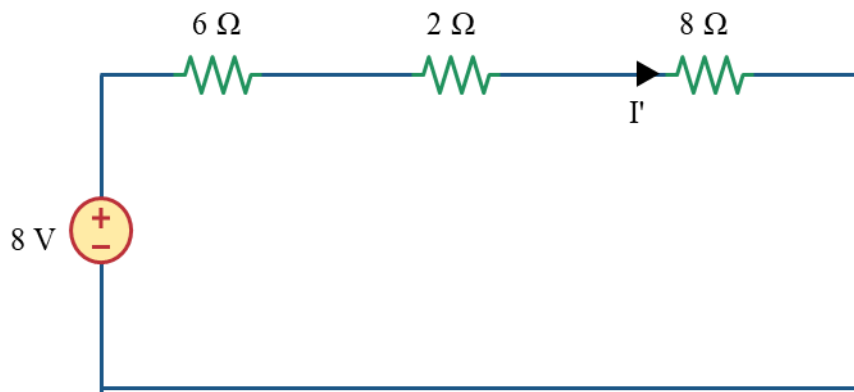
Solution to Problem 2



If I' , I'' and I''' are the contributions from the 8 V voltage source, 2 A current source and 6 V voltage source respectively, then

$$I = I' + I'' + I'''$$

First, Considering the contribution of the 8 V voltage source (replacing the current source (2 A) by an open circuit and voltage source (6 V) by a short circuit)

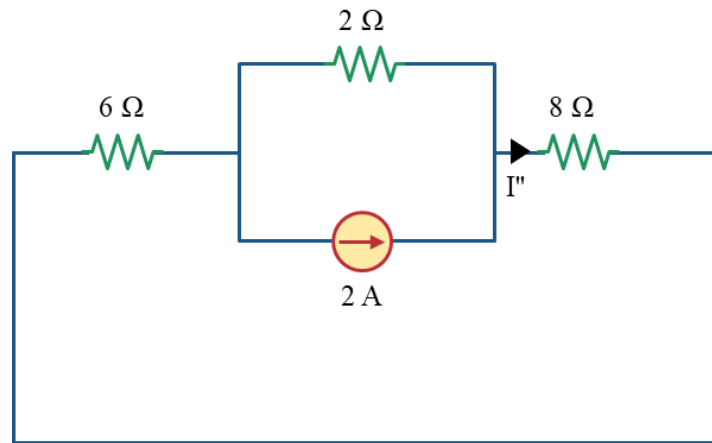


Applying KVL in the circuit,

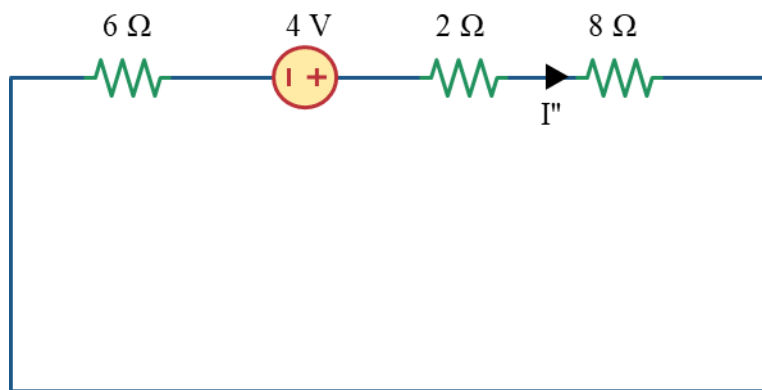
$$-8 + 6I' + 2I' + 8I' = 0$$

$$I' = 0.5 \text{ A}$$

Considering the contribution of the 2A current source (replacing the voltage sources (8 V & 6 V) by a short circuits)



Converting the current source into voltage source to simplify the circuit,

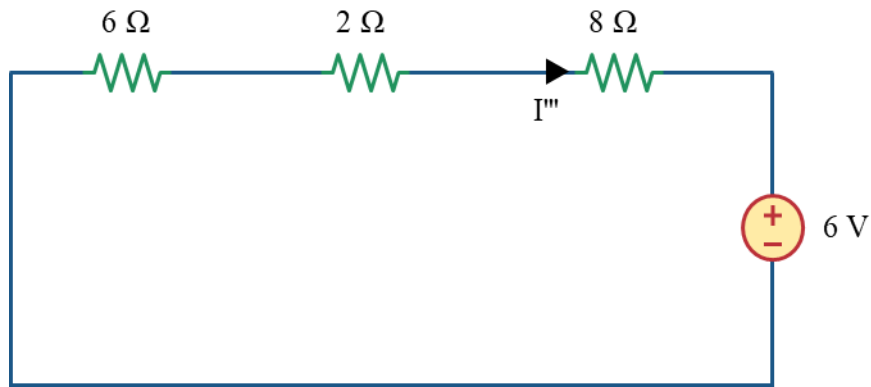


Applying KVL in the circuit,

$$-4 + 2I'' + 8I'' + 6I'' = 0$$

$$I'' = 0.25 \text{ A}$$

Considering the contribution of the 6 V voltage source (replacing the current source (2 A) by an open circuit and voltage source (8 V) by a short circuit)



Applying KVL in the circuit,

$$6 + 6I''' + 2I''' + 8I''' = 0$$

$$I''' = -0.375 \text{ A}$$

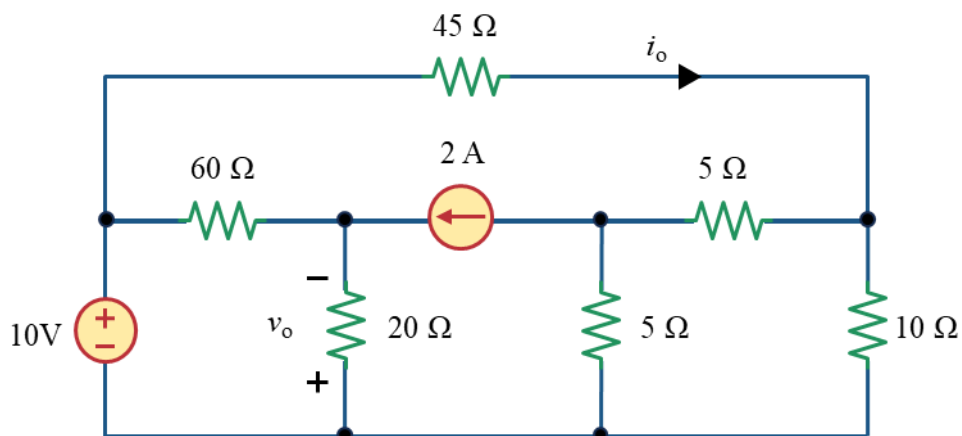
Finally,

$$I = I' + I'' + I'''$$

$$I = 0.5 + 0.25 - 0.375$$

$$I = 0.375 \text{ A}$$

Solution to Problem 3

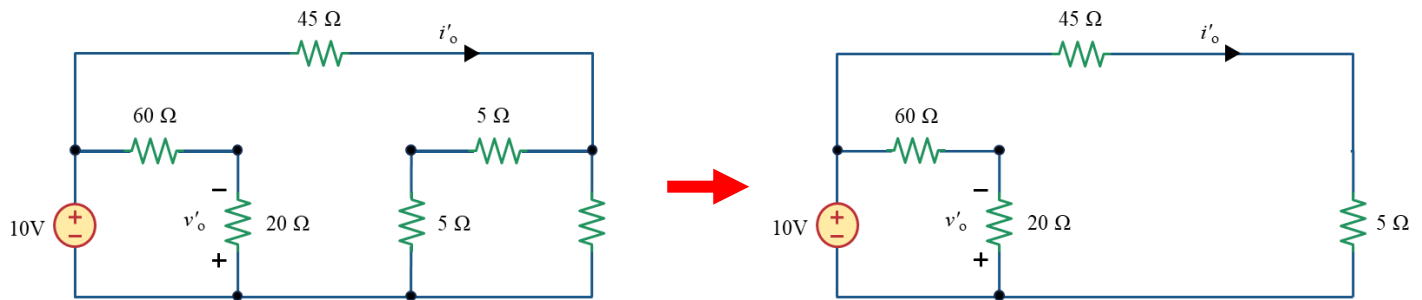


If v_0' (and i_0') and v_0'' (and i_0'') are the contributions from the 10 V voltage source and 2 A current source respectively, then

$$v_0 = v_0' + v_0''$$

$$i_0 = i_0' + i_0''$$

Considering the contribution of the 10 V voltage source (replacing the current source (2 A) by an open circuit). The resistors in the right portion have been replaced by equivalent resistance, $(5 + 5) || 10 = 5\Omega$



As, 10 V is across the $(45 + 5) = 50 \Omega$, the current i'_0 through it will be,

$$i'_0 = \frac{10}{50} = 0.2 \text{ A}$$

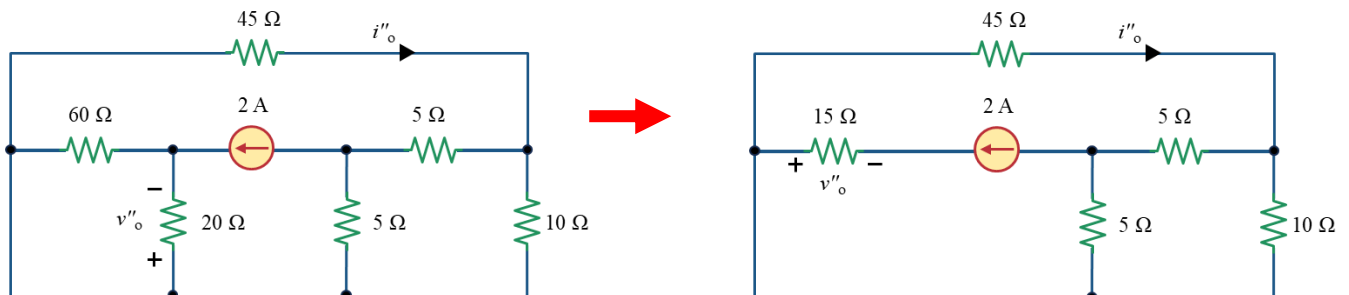
As, 10 V is across the $(60 + 20) = 80 \Omega$, the voltage v'_0 across 20Ω will be, (Using voltage divider rule)

$$v'_0 = 10 \times \frac{-20}{80} = -2.5 \text{ V}$$

The – (minus) sign is due to given polarity of v_0

Now, considering the contribution of the 2 A current source (replacing the voltage source (10 V) by a short circuit). Let's find out v_0'' first

We can see that; the 2 A current is divided into the 20Ω and 60Ω resistors. Infact, if we look closely, the voltage v_0'' is not only across 20Ω , but also across 60Ω . So, if we find the equivalent resistance $(20 \parallel 60 = 15 \Omega)$ of these resistors then we can find the v_0'' by multiplying the current with the equivalent resistance. In that case, the circuit will be,

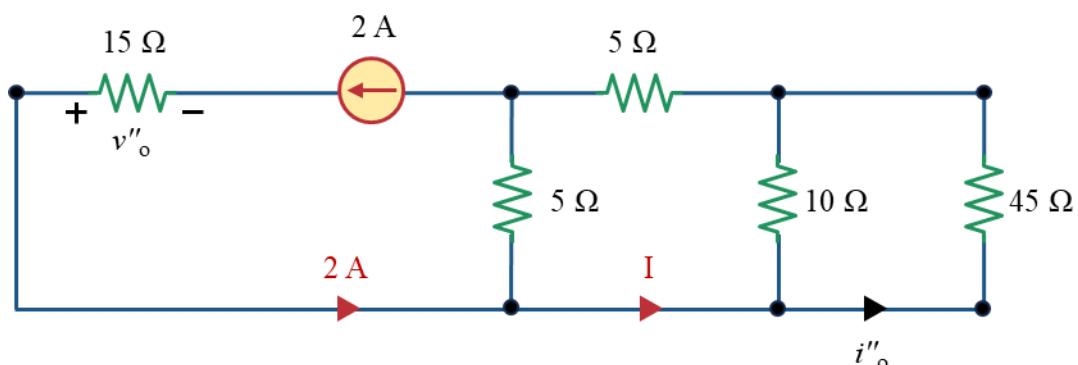


$$v''_0 = -2 \times 15 = -30 \text{ V}$$

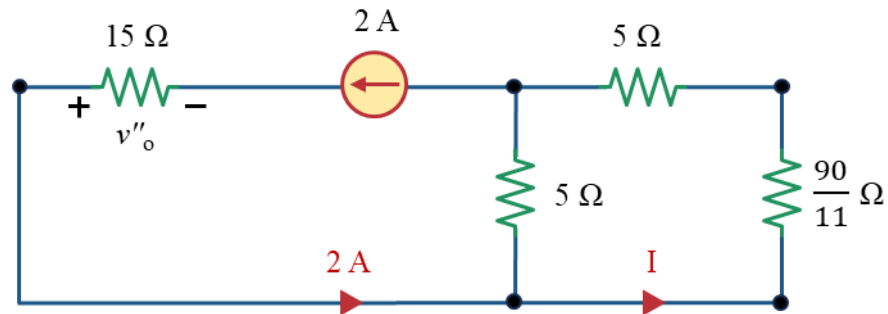
The – (minus) sign is due to given polarity of v_0

Let's find out i_0''

If we look closely we can see that the 45Ω is parallel with 10Ω . We can rearrange the circuit as follows,



To find out the i_0'' , we first have to calculate I. To calculate we need to have equivalent resistance of the parallel $45\ \Omega$ and $10\ \Omega$ ($45 \parallel 10 = \frac{90}{11}\ \Omega$). So, the circuit becomes



The equivalent resistance on the right portion:

$$R_{eq} = \left(\left(\frac{90}{11} + 5 \right) \parallel 5 \right) = \frac{29}{8}$$

Using current divider rule,

$$I = 2 \times \left(\frac{\frac{29}{8}}{\frac{90}{11} + 5} \right) = 0.55\text{ A}$$

Now,

$$i_0'' = I \times \frac{90}{45}$$

$$i_0'' = 0.55 \times \frac{90}{45}$$

$$i_0'' = 0.1\text{ A}$$

Finally,

$$v_0 = v_0' + v_0''$$

$$v_0 = -2.5 - 30$$

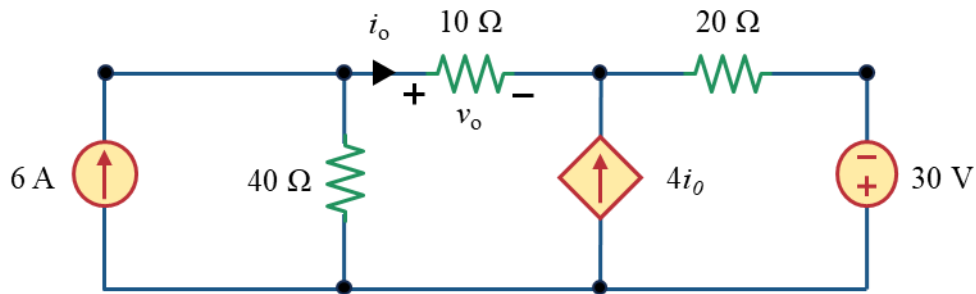
$$v_0 = -32.5\text{ V}$$

$$i_0 = i_0' + i_0''$$

$$i_0 = 0.2 + 0.1$$

$$i_0 = 0.3\text{ A}$$

Solution to Problem 4



If v_o' (and i_o') and v_o'' (and i_o'') are the contributions from the 6 A current source and 30 V voltage source respectively, then

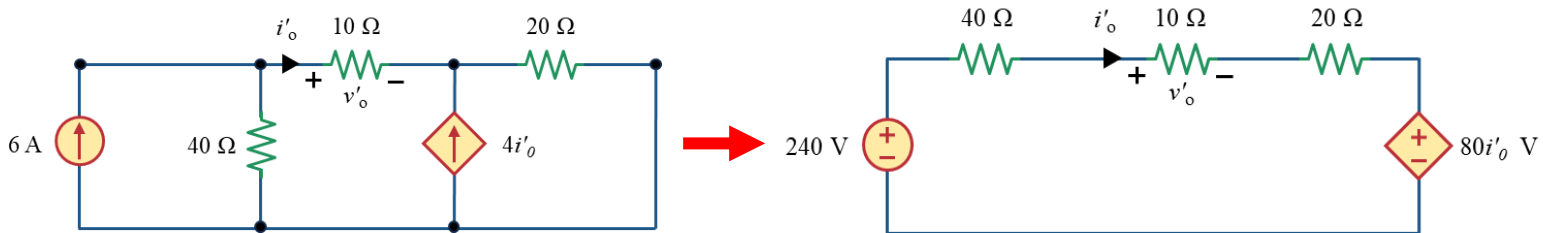
$$v_o = v_o' + v_o''$$

$$i_o = i_o' + i_o''$$

Now, considering the contribution of the 6 A current source (replacing the voltage source (30 V) by a short circuit).

If we look closely, we can apply source transformation for 6 A current source and $4i_o'$ dependent current source and convert them into voltage sources with resistance in series. This will simplify the circuit

Could have used Nodal Analysis



Applying KVL in the circuit,

$$-240 + 40 i_o' + 10 i_o' + 20 i_o' + 80 i_o' = 0$$

$$i_o' = 1.6 \text{ A}$$

Now,

$$v_o' = 10 i_o'$$

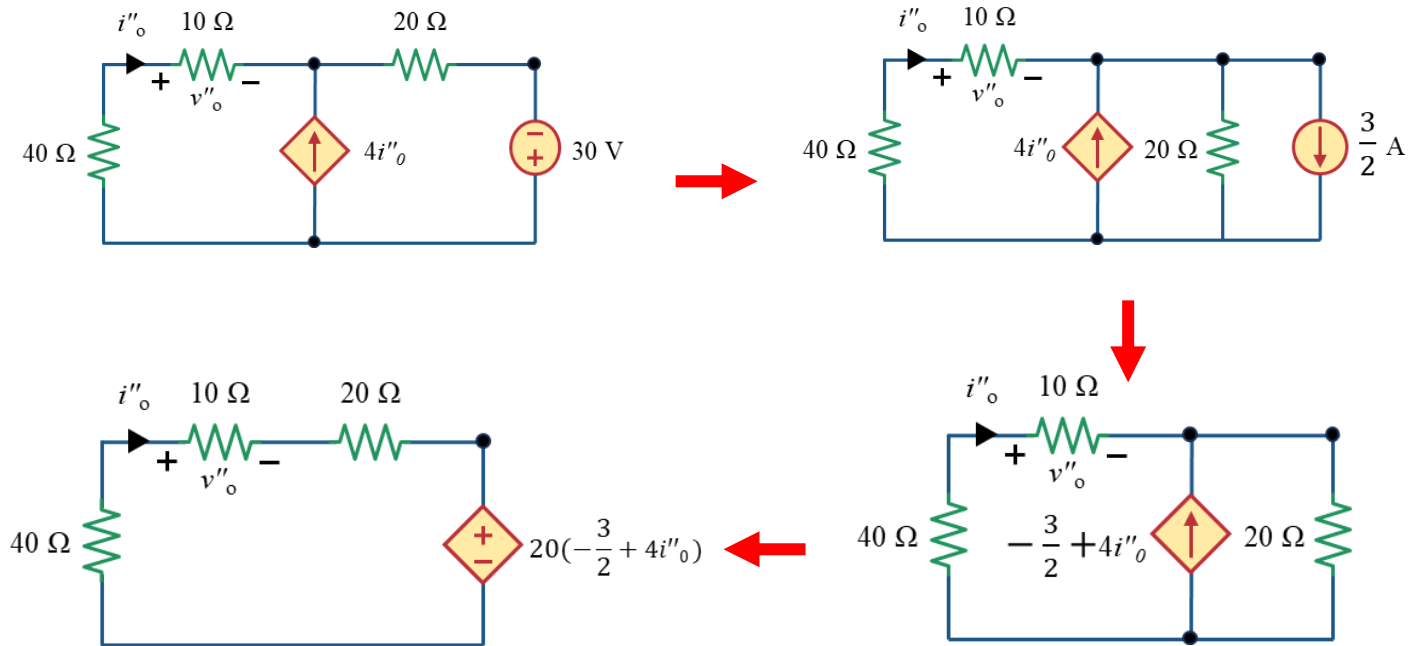
$$v_o' = 10 \times 1.6$$

$$v_o' = 16 \text{ V}$$

Now, considering the contribution of the 30 V voltage source (replacing the current source (6 A) by an open circuit).

Now, if we do a series of source transformations as in the figures below, the circuit will become simple.

Could have used Mesh Analysis



Applying KVL in the circuit,

$$40i''_0 + 10i''_0 + 20i''_0 + 20\left(-\frac{3}{2} + 4i''_0\right) = 0$$

$$40i''_0 + 10i''_0 + 20i''_0 - \frac{40}{3} + 80i''_0 = 0$$

$$i''_0 = 0.2 \text{ A}$$

Now,

$$v''_0 = 10i''_0$$

$$v''_0 = 10 \times 0.2$$

$$v''_0 = 2 \text{ V}$$

Finally,

$$v_0 = v'_0 + v''_0$$

$$v_0 = 16 + 2$$

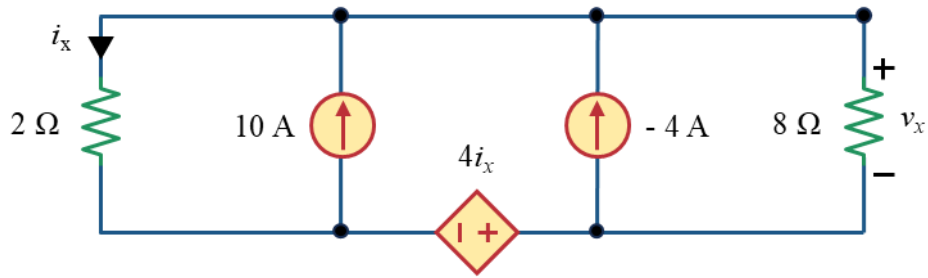
$$v_0 = 18 \text{ V}$$

$$i_0 = i'_0 + i''_0$$

$$i_0 = 1.6 + 0.2$$

$$i_0 = 1.8 \text{ A}$$

Solution to Problem 5

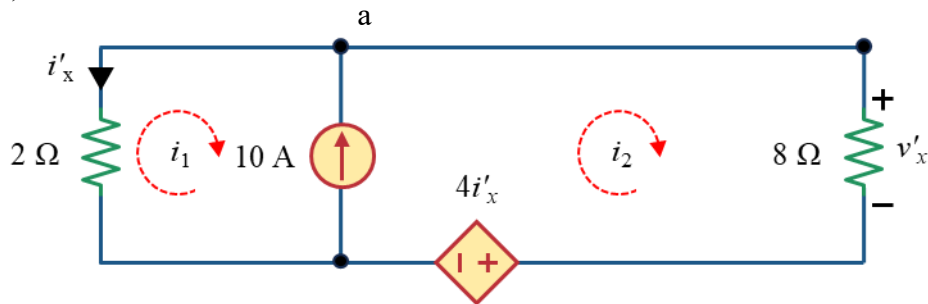


If v_0' (and i_0') and v_0'' (and i_0'') are the contributions from the 10 A current source and - 4 A current source respectively, then

$$v_x = v_x' + v_x''$$

$$i_x = i_x' + i_x''$$

Now, considering the contribution of the 10 A current source (replacing the current source (- 4 A) by an open circuit)



To find the required variables we can apply mesh analysis in the circuit. From the circuit,

$$i_x' = -i_1$$

$$v_x' = 8i_2$$

We can see that there is supermesh. Applying KVL in supermesh,

$$2i_1 + 8i_2 + 4i_x' = 0$$

$$2i_1 + 8i_2 + 4(-i_1) = 0$$

$$-2i_1 + 8i_2 = 0 \quad \dots\dots\dots (i)$$

Applying KCL in node a,

$$i_1 - i_2 = -10 \quad \dots\dots\dots (ii)$$

Solving (i) and (ii),

$$i_1 = -\frac{40}{3}$$

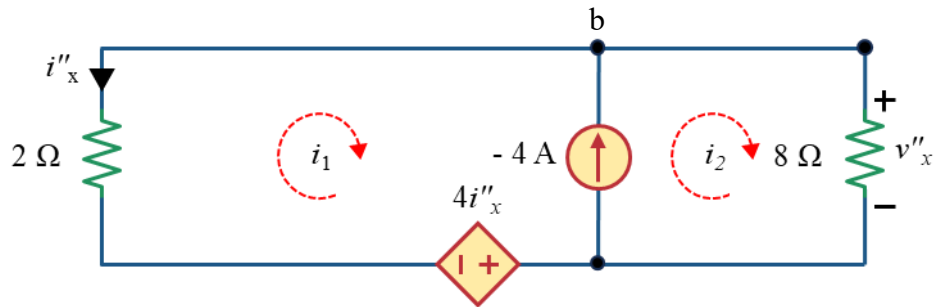
$$i_2 = -\frac{10}{3}$$

Now,

$$i'_x = -\left(-\frac{40}{3}\right) = \frac{40}{3} \text{ A}$$

$$v'_x = -\frac{10}{3} \times 8 = -\frac{80}{3} \text{ V}$$

Considering the contribution of the - 4 A current source (replacing the current source (10 A) by an open circuit)



To find the required variables we can apply mesh analysis in the circuit. From the circuit,

$$i''_x = -i_1$$

$$v''_x = 8i_2$$

We can see that there is supermesh. Applying KVL in supermesh,

$$2i_1 + 8i_2 + 4i''_x = 0$$

$$2i_1 + 8i_2 + 4(-i_1) = 0$$

$$-2i_1 + 8i_2 = 0 \quad \dots\dots\dots (i)$$

Applying KCL in node b,

$$i_1 - i_2 = 4 \quad \dots\dots\dots (ii)$$

Solving (i) and (ii),

$$i_1 = \frac{16}{3}$$

$$i_2 = \frac{4}{3}$$

Now,

$$i''_x = -\left(\frac{16}{3}\right) = -\frac{16}{3} \text{ A}$$

$$v''_x = \frac{4}{3} \times 8 = \frac{32}{3} \text{ V}$$

Finally,

$$v_0 = v'_0 + v''_0$$

$$v_0 = -\frac{80}{3} + \frac{32}{3}$$

$$v_0 = -16 \text{ V}$$

$$i_0 = i'_0 + i''_0$$

$$i_0 = \frac{40}{3} + \frac{32}{3}$$

$$i_0 = 24 \text{ A}$$